MIL-T-13644(Ord)
15 September 1954
SUPERSEDING
Army 51-72-42
9 September 1954

MILITARY SPECIFICATION

TELESCOPE, ELBOW, M35

1. SCOPE

1.1 This specification covers an elbow telescope used as the battery commander's observation telescope on the Instrument, Observation, A.A., B.C., M1.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

U. S. Army
- 51-7 - Instruments, Fire Control; General Specification Governing the Manufacture and Inspection of
- 51-70-2 - Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Unless otherwise specified, the requirements common to all fire control instruments as contained in Specifications 51-7 and 51-70-2 shall be mandatory.

3.2 Material.-Materials shall be as specified in the applicable drawings and specifications forming a part of this specification.

3.3 Eyepiece focus.-The diopter scale shall indicate zero reading within ± 0.25 diopter when the reticle line at the center of the field is brought into sharp focus with an auxiliary telescope of at least 3 power. The diopter scale shall have sufficient movement to cover the range from plus 3 to minus 3 diopters.

3.4 Collimation.-The optical axis, denoted by the center of the reticle, shall be perpendicular to the longitudinal axis of the trunnion bearings within a tolerance of 0.5 mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:58Z
Check the source to verify that this is the current version before use.

3.5 Parallax.-With the target as 500 yards plus or minus 50 yards, the parallax in the center of the field shall not exceed 0.05 mil (approximately 1 reticle width).

3.6 Resolution.-Resolving power on the optical axis for a parallel-bar test target shall be 2.5 seconds of arc or better.

3.7 Filters.-When positioned in the optical path, filters shall not affect telescope definition.

3.8 Plumb of reticle.-With the geometric axis of the telescope in a true horizontal plane and the telescope set for 10-power, the vertical scale of the reticle shall not deviate from the image of a plumb line by more than the angle subtended by a 0.5 mil block at the inner end of one 50-mil graduation when the inner end of the other 50-mil graduation is in coincidence with the plumb line. This check may be performed using the horizontal reticle scale and a level line.

3.9 Reticle accuracy.-The accuracy of any graduation shall be within 2 percent of the indicated value plus or minus 0.08 mil.

3.10 Image tilt.-The image of a plumb line shall be within one degree of the vertical as measured by an auxiliary telescope in back of the eyepiece.

3.11 Vibration.-After being subjected to the vibration test specified in Specification 51-70-2, except that 2 to 2-1/2 minutes shall be substituted for 15 to 18 seconds, there shall be no damage to the instrument, no change in collimation in excess of 0.2 mil, and no loose dirt in excess of the following:

3.11.1 Dirt particles on the adjacent surfaces of the objective and the prism shall not exceed a total of 6 in number of a size not larger than a 50 dig.

3.11.2 Dirt particles on the adjacent surfaces of the prism and the reticle shall not exceed a total of 4 in number of a size not larger than a 2 dig.

3.11.3 Dirt particles on the adjacent surfaces of the reticle collective lens and the first erector shall not exceed a total of 3 in number of a size not larger than a 5 dig.

3.11.4 Dirt particles on the adjacent surfaces of the first and second erectors shall not exceed a total of 2 in number of a size not larger than a 50 dig.

3.11.5 Dirt particles on the adjacent surfaces of the second erector and any filter element shall not exceed a total of 3 in number of a size not larger than a 5 dig.

3.11.6 Dirt particles on the adjacent surfaces of any filter element and the field lens shall not exceed a total of 4 in number of a size not larger than a 2 dig.

3.11.7 Dirt particles on the adjacent surfaces of the field lens and center lens shall not exceed a total of 4 in number of a size not larger than a 20 dig.

3.11.8 Dirt particles on the adjacent surfaces of the center lens and eye-lens shall not exceed a total of 4 in number of a size not larger than a 20 dig.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:58Z
Check the source to verify that this is the current version before use.

3.12 Rain.-There shall be no entrance of water into the telescope when it is subjected to the rain test specified in 4.3.

3.13 Temperature.-The telescope shall meet the temperature requirements for Type I instruments specified in Specification 51-7. To be mechanically operable, the diopter adjustment shall turn through its full range when subjected to an applied torque of 60 inch-pounds, and the filter selector shall move under an applied torque not to exceed 10 inch-pounds on the long lever and 6.5 inch-pounds on the short lever.

3.14 Illumination.-The reticle markings shall appear clearly defined when illuminated in the specified manner and observed in a dark room.

3.15 Workmanship.-Workmanship shall be in accordance with Specifications 51-7 and 51-70-2.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.-Unless otherwise specified, each telescope shall be inspected for conformance with this specification and with Specifications 51-7 and 51-70-2.

4.2 Definition test.-The test object shall be of the pattern illustrated in Specification 51-70-2, Figure VI-A. The image shall be viewed with the aid of a 6-power auxiliary telescope of approved design, scaled in diopters. The image of the test pattern shall show correct line count, and the pattern shall be resolved in each of the four meridians by an adjustment not to exceed ± 1/8 diopter from the setting for the best focus of the numeral in the center of the test field.

4.3 Rain test.-The water and the telescope shall be at room temperature during the test. The shower produced shall be a steady downpour simulating a rainfall of 4.0 inches per hour, measured at the surface of the instrument. The spray shall be directed on the instrument for 2 hours in directions ranging from vertical to 30° from the vertical. The simulated rain shall be directed on all surfaces of the telescope normally exposed to rain when the instrument is in its operating position.

4.4 Temperature test.-The temperature test shall be performed as specified in Specification 51-7.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with instructions forming part of the contract, or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, date of this specification, and the type of packing.

Patent Notice: This specification, together with specifications, and drawings pertaining to it, and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:58Z
Check the source to verify that this is the current version before use.

when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Custodian:

Army-Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:58Z
Check the source to verify that this is the current version before use.